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9 **(a)** Simplify.

$$4x + 3y + 2x - 8y$$

..... [2]

(b) A pen costs 60 cents and a ruler costs 29 cents.

Write down an expression for the total cost, in cents, of x pens and y rulers.

..... cents [2]

(c) Solve.

$$5(2x + 4) = 85$$

$x =$ ..... [3]

(d) **(i)** $2^8 \times 2^m = 2^6$

Work out the value of m.

$m =$ ..... [1]

(ii) $5^n \div 5^4 = 5^6$

Work out the value of n.

$n =$ ..... [1]

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(e) A plant costs p dollars and a bush costs b dollars.
Ana buys 2 plants and 4 bushes for \$42.
Paola buys 7 plants and 9 bushes for \$107.

Write down a pair of simultaneous equations and solve them to find the value of p and the value of b.
You must show all your working.

p = ...

b = ... [6]